MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-BVOC (HONOURS) Regular EXAMINATION November 2025

FIRST SEMESTER

Multi-Disciplinary Courses (MDC) - MG1MDCFTA100 - FOOD PROCESSING TECHNOLOGY
(2025 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Part A

One or two Sentences
Answer any 10 questions
Each question carries 1 marks

1. Advancements: automation, refrigeration, UHT, freeze-drying, improved packaging.

2. Vitamin C reduces due to heat-sensitive degradation during processing.

3. Cryogenic freezing uses extremely low temperatures like liquid nitrogen (–196°C).

4. Sugar >65% solids; salt 15–20% inhibits microbial growth.

5. Irradiation: Radurization, Radicidation, Radappertization.

6. Sterilization temperature range: 110–121°C.

7. Prevent sugar bloom by proper tempering and stable storage temperature.

8. Pickling methods: Salt pickling, Vinegar pickling, Fermented pickling.

9. Sauce defect: Syneresis; prevented by correct starch use and cooking.

10. Roasting peanuts improves flavour, colour, and reduces moisture.

11. Canned soups sterilized at 121°C for 15–20 minutes.

12. Cloudy jelly occurs due to improper straining or excess pectin.

Part B

Short answer
Answer any 4 questions
Each question carries 5 marks

13. Benefits & limitations of secondary processing:

Benefits: longer shelf life, improved sensory quality, value addition, reduced waste, convenience.
Limitations: nutrient loss, need for equipment, higher cost, contamination risk.

14. Objectives & methods of conventional cooking:

Objectives: improve palatability, digestibility, safety, flavour.

Methods: boiling, steaming, baking, roasting, frying, pressure cooking.

15. LTLT, HTST & UHT pasteurization:

LTLT: 63°C for 30 min.
HTST: 72°C for 15 sec.
UHT: 135–150°C for 2–5 sec.

16. Principle of high-temperature preservation:

Destroys microorganisms, denatures enzymes, achieves commercial sterility.
Used in blanching, pasteurization, sterilization.

17. Packaging & storage requirements for ketchup:

Use glass/plastic containers, hot-fill, proper sealing.
Store in cool dry place; refrigerate after opening.

18. Sugar–pectin–acid balance in jam:

Affects gel formation, prevents syneresis/crystallization.
Ideal: pH 3.0–3.3, 65% TSS, correct pectin level.

Part C

Essay Questions
Answer any 2 questions
Each question carries 10 marks

19. Canning process:

Selection & grading, washing, peeling, cutting, blanching.
Filling, exhausting, sealing, sterilization at 110–121°C, cooling, storage.
Safety: hygiene, correct time–temperature, avoid under-processing.

20. Refrigeration – principles, types, limitations:

Principle: slows microbial growth by lowering temperature.
Types: natural, mechanical, cryogenic.
Limitations: energy use, does not kill microbes, temperature fluctuations.

21. Contribution of low-temperature preservation:

Extends shelf life, retains nutrients and colour, reduces waste.
Supports frozen/chilled food industry and global supply chains.

22. Marmalade production:

Washing, peeling, juice extraction, peel cutting.
Boiling with sugar, concentrating to 65% TSS, adding peel.
Testing endpoint, filling hot, cooling, sealing.